

Deep Neural Networks - Programming Assignment

Comparing Linear Models and Multi-Layer Perceptrons

Student Name: Sumanth N

Student ID: 2025AC05403

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IMPORTANT INSTRUCTIONS

1. **Complete ALL sections** marked with `TODO`
 2. **DO NOT modify** the `get_assignment_results()` function structure
 3. **Fill in all values accurately** - these will be auto-verified
 4. **After submission**, you'll receive a verification quiz based on YOUR results
 5. **Run all cells** before submitting (Kernel → Restart & Run All)
-

```
In [1]: # Import required libraries
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
import time
import warnings
warnings.filterwarnings('ignore')

# Set random seed for reproducibility
np.random.seed(42)
print('✓ Libraries imported successfully')
```

✓ Libraries imported successfully

Section 1: Dataset Selection and Loading

Requirements:

- ≥ 500 samples
- ≥ 5 features
- Public dataset (UCI/Kaggle)
- Regression OR Classification problem

```
In [2]: # Load dataset
data_path = "heart.csv"
data = pd.read_csv(data_path)

# Dataset information
dataset_name = "Heart Failure Prediction"
dataset_source = "Kaggle (fedesoriano)"
n_samples = int(data.shape[0]) # 918
n_features = int(data.shape[1] - 1) # 11 (excluding target)
problem_type = "binary_classification"

# Problem statement
problem_statement = """
Predicting the presence of heart disease based on 11 clinical features (e.g.
Early prediction of cardiovascular disease is vital in clinical medicine to
"""

# Primary evaluation metric
primary_metric = "accuracy"

# Metric justification
metric_justification = """
We use classification accuracy as our primary metric to evaluate the overall
Additionally, we compute precision, recall, and F1-score to analyze performance
"""

print(f"Dataset: {dataset_name}")
print(f"Source: {dataset_source}")
print(f"Samples: {n_samples}, Features: {n_features}")
print(f"Problem Type: {problem_type}")
print(f"Primary Metric: {primary_metric}")
```

Dataset: Heart Failure Prediction
Source: Kaggle (fedesoriano)
Samples: 918, Features: 11
Problem Type: binary_classification
Primary Metric: accuracy

Section 2: Data Preprocessing

Preprocess your data:

1. Handle missing values
2. Encode categorical variables
3. Split into train/test sets
4. Scale features

```
In [3]: # Preprocess your data
# 1. Separate features (X) and target (y)
X = data.drop('HeartDisease', axis=1)
y = data['HeartDisease']

# 2. One-hot encode categorical variables
```

```

X_encoded = pd.get_dummies(X, columns=['Sex', 'ChestPainType', 'RestingECG'],
X_encoded = X_encoded.astype(float)

# Train-test split (80-20 ratio)
X_train, X_test, y_train, y_test = train_test_split(X_encoded, y, test_size=

# Feature scaling using StandardScaler
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

# Fill these after preprocessing
train_samples = int(X_train.shape[0])          # 734
test_samples = int(X_test.shape[0])            # 184
train_test_ratio = float(train_samples / (train_samples + test_samples)) #

print(f"Train samples: {train_samples}")
print(f"Test samples: {test_samples}")
print(f"Split ratio: {train_test_ratio:.1%}")

```

Train samples: 734

Test samples: 184

Split ratio: 80.0%

Section 3: Baseline Model Implementation

Implement from scratch (NO sklearn models!):

- Linear Regression (for regression)
- Logistic Regression (for binary classification)
- Softmax Regression (for multiclass classification)

Must include:

- Forward pass (prediction)
- Loss computation
- Gradient computation
- Gradient descent loop
- Loss tracking

```

In [4]: class BaselineModel:
        """
        Baseline linear model with gradient descent
        Implement: Logistic Regression for binary classification
        """
        def __init__(self, learning_rate=0.1, n_iterations=3000):
            self.lr = learning_rate
            self.n_iterations = n_iterations
            self.weights = None
            self.bias = None
            self.loss_history = []

```

```

def sigmoid(self, z):
    return 1 / (1 + np.exp(-np.clip(z, -500, 500)))

def fit(self, X, y):
    """
    Implement gradient descent training
    """
    n_samples, n_features = X.shape
    self.weights = np.zeros(n_features)
    self.bias = 0.0

    y_np = y.values if hasattr(y, 'values') else y

    for i in range(self.n_iterations):
        # 1. Forward pass: predictions
        linear_output = np.dot(X, self.weights) + self.bias
        y_pred = self.sigmoid(linear_output)

        # 2. Compute loss (binary cross entropy)
        epsilon = 1e-15
        y_pred = np.clip(y_pred, epsilon, 1.0 - epsilon)
        loss = -np.mean(y_np * np.log(y_pred) + (1.0 - y_np) * np.log(1.
        self.loss_history.append(loss)

        # 3. Compute gradients
        dw = (1 / n_samples) * np.dot(X.T, (y_pred - y_np))
        db = (1 / n_samples) * np.sum(y_pred - y_np)

        # 4. Update weights and bias
        self.weights -= self.lr * dw
        self.bias -= self.lr * db

    return self

def predict(self, X):
    """
    Implement prediction (binary labels 0 or 1)
    """
    linear_output = np.dot(X, self.weights) + self.bias
    y_pred = self.sigmoid(linear_output)
    return (y_pred >= 0.5).astype(int)

print("✓ Baseline model class defined")

```

✓ Baseline model class defined

```

In [5]: # Train baseline model
print("Training baseline model...")
baseline_start_time = time.time()

# Initialize and train baseline model
baseline_model = BaselineModel(learning_rate=0.1, n_iterations=3000)
baseline_model.fit(X_train_scaled, y_train)

# Make predictions
baseline_predictions = baseline_model.predict(X_test_scaled)

```

```
baseline_training_time = time.time() - baseline_start_time
print(f"✓ Baseline training completed in {baseline_training_time:.4f}s")
print(f"✓ Loss decreased from {baseline_model.loss_history[0]:.4f} to {baseline_model.loss_history[-1]:.4f}")
```

Training baseline model...

✓ Baseline training completed in 0.1329s

✓ Loss decreased from 0.6931 to 0.3202

Section 4: Multi-Layer Perceptron Implementation

Implement MLP from scratch with:

- At least 1 hidden layer
- ReLU activation for hidden layers
- Appropriate output activation
- Forward propagation
- Backward propagation
- Gradient descent

```
In [6]: class MLP:
        """
        Multi-Layer Perceptron implemented from scratch
        """
        def __init__(self, architecture, learning_rate=0.1, n_iterations=3000, beta=0.9):
            """
            architecture: list [input_size, hidden1, hidden2, ..., output_size]
            """
            self.architecture = architecture
            self.lr = learning_rate
            self.n_iterations = n_iterations
            self.beta = beta
            self.parameters = {}
            self.v = {} # velocity for momentum
            self.loss_history = []
            self.cache = {}

            def initialize_parameters(self):
                """
                Initialize weights and biases for all layers
                """
                np.random.seed(42)

                for l in range(1, len(self.architecture)):
                    input_size = self.architecture[l-1]
                    output_size = self.architecture[l]
                    # He initialization for ReLU layers
                    self.parameters[f'W{l}'] = np.random.randn(output_size, input_size)
                    self.parameters[f'b{l}'] = np.zeros((output_size, 1))
                    # Initialize velocities for momentum
                    self.v[f'dW{l}'] = np.zeros_like(self.parameters[f'W{l}'])
                    self.v[f'db{l}'] = np.zeros_like(self.parameters[f'b{l}'])
```

```

def relu(self, Z):
    """ReLU activation function"""
    return np.maximum(0, Z)

def relu_derivative(self, Z):
    """ReLU derivative"""
    return (Z > 0).astype(float)

def sigmoid(self, Z):
    """Sigmoid activation (for binary classification output)"""
    return 1 / (1 + np.exp(-np.clip(Z, -500, 500)))

def forward_propagation(self, X):
    """
    Implement forward pass through all layers
    """
    # Transpose X to shape (n_features, m_samples)
    A = X.T
    self.cache['A0'] = A
    L = len(self.architecture) - 1

    for l in range(1, L):
        W = self.parameters[f'W{l}']
        b = self.parameters[f'b{l}']
        Z = np.dot(W, A) + b
        A = self.relu(Z)
        self.cache[f'Z{l}'] = Z
        self.cache[f'A{l}'] = A

    # Output layer (sigmoid)
    W = self.parameters[f'W{L}']
    b = self.parameters[f'b{L}']
    Z = np.dot(W, A) + b
    A_out = self.sigmoid(Z)
    self.cache[f'Z{L}'] = Z
    self.cache[f'A{L}'] = A_out

    return A_out

def backward_propagation(self, X, y):
    """
    Implement backward pass to compute gradients
    """
    m = X.shape[0]
    grads = {}
    L = len(self.architecture) - 1

    # Reshape y to (1, m)
    y_reshaped = y.values.reshape(1, -1) if hasattr(y, 'values') else y

    # Output layer dZ
    A_out = self.cache[f'A{L}']
    dZ = A_out - y_reshaped

    for l in reversed(range(1, L + 1)):
        A_prev = self.cache[f'A{l-1}']

```

```

        grads[f'dW{l}'] = (1.0 / m) * np.dot(dZ, A_prev.T)
        grads[f'db{l}'] = (1.0 / m) * np.sum(dZ, axis=1, keepdims=True)
        if l > 1:
            W = self.parameters[f'W{l}']
            dZ = np.dot(W.T, dZ) * self.relu_derivative(self.cache[f'Z{l}'])

    return grads

def update_parameters(self, grads):
    """
    Update weights and biases using gradients with Momentum
    """
    L = len(self.architecture) - 1
    for l in range(1, L + 1):
        self.v[f'dW{l}'] = self.beta * self.v[f'dW{l}'] + (1 - self.beta) * grads[f'dW{l}']
        self.v[f'db{l}'] = self.beta * self.v[f'db{l}'] + (1 - self.beta) * grads[f'db{l}']
        self.parameters[f'W{l}'] -= self.lr * self.v[f'dW{l}']
        self.parameters[f'b{l}'] -= self.lr * self.v[f'db{l}']

def compute_loss(self, y_pred, y_true):
    """
    Compute loss (binary cross-entropy)
    """
    epsilon = 1e-15
    y_pred = np.clip(y_pred, epsilon, 1.0 - epsilon)
    y_resaped = y_true.values.reshape(1, -1) if hasattr(y_true, 'values') else y_true
    loss = -np.mean(y_resaped * np.log(y_pred) + (1.0 - y_resaped) * np.log(1 - y_pred))
    return loss

def fit(self, X, y):
    """
    Implement training loop with learning rate decay
    """
    self.initialize_parameters()

    for i in range(self.n_iterations):
        if i > 0 and i % 1000 == 0:
            self.lr *= 0.5
        # 1. Forward propagation
        y_pred = self.forward_propagation(X)

        # 2. Compute loss
        loss = self.compute_loss(y_pred, y)
        self.loss_history.append(loss)

        # 3. Backward propagation
        grads = self.backward_propagation(X, y)

        # 4. Update parameters
        self.update_parameters(grads)

    return self

def predict(self, X):
    """
    Implement prediction (threshold at 0.5)
    """

```

```

        """
        y_pred = self.forward_propagation(X)
        return (y_pred >= 0.5).astype(int).flatten()

print("✓ MLP class defined")

```

✓ MLP class defined

```

In [7]: # Train MLP
print("Training MLP...")
mlp_start_time = time.time()

# Define your architecture and train MLP
n_features = X_train_scaled.shape[1]
mlp_architecture = [n_features, 16, 8, 1]
mlp_model = MLP(architecture=mlp_architecture, learning_rate=0.1, n_iteratic
mlp_model.fit(X_train_scaled, y_train)

# Make predictions
mlp_predictions = mlp_model.predict(X_test_scaled)

mlp_training_time = time.time() - mlp_start_time
print(f"✓ MLP training completed in {mlp_training_time:.4f}s")
print(f"✓ Loss decreased from {mlp_model.loss_history[0]:.4f} to {mlp_model.

```

Training MLP...

✓ MLP training completed in 0.5673s

✓ Loss decreased from 0.7610 to 0.1736

Section 5: Evaluation and Metrics

Calculate appropriate metrics for your problem type

```

In [8]: def calculate_metrics(y_true, y_pred, problem_type):
        """
        Calculate appropriate metrics based on problem type
        """
        metrics = {}
        y_true_np = y_true.values if hasattr(y_true, 'values') else y_true
        y_pred_np = y_pred.values if hasattr(y_pred, 'values') else y_pred

        if problem_type == "regression":
            pass
        elif problem_type in ["binary_classification", "multiclass_classification"]
            tp = np.sum((y_true_np == 1) & (y_pred_np == 1))
            tn = np.sum((y_true_np == 0) & (y_pred_np == 0))
            fp = np.sum((y_true_np == 0) & (y_pred_np == 1))
            fn = np.sum((y_true_np == 1) & (y_pred_np == 0))

            accuracy = (tp + tn) / len(y_true_np)
            precision = tp / (tp + fp) if (tp + fp) > 0 else 0.0
            recall = tp / (tp + fn) if (tp + fn) > 0 else 0.0
            f1 = 2 * precision * recall / (precision + recall) if (precision + r

        metrics['test_accuracy'] = float(accuracy)

```



```

        metrics['test_precision'] = float(precision)
        metrics['test_recall'] = float(recall)
        metrics['test_f1'] = float(f1)

    return metrics

# Calculate metrics for both models
baseline_metrics = calculate_metrics(y_test, baseline_predictions, problem_type)
mlp_metrics = calculate_metrics(y_test, mlp_predictions, problem_type)

print("Baseline Model Performance:")
print(baseline_metrics)

print("\nMLP Model Performance:")
print(mlp_metrics)

```

Baseline Model Performance:

```
{'test_accuracy': 0.8532608695652174, 'test_precision': 0.9, 'test_recall': 0.8411214953271028, 'test_f1': 0.8695652173913043}
```

MLP Model Performance:

```
{'test_accuracy': 0.8913043478260869, 'test_precision': 0.9065420560747663, 'test_recall': 0.9065420560747663, 'test_f1': 0.9065420560747663}
```

Section 6: Visualization

Create visualizations:

1. Training loss curves
2. Performance comparison
3. Additional domain-specific plots

```

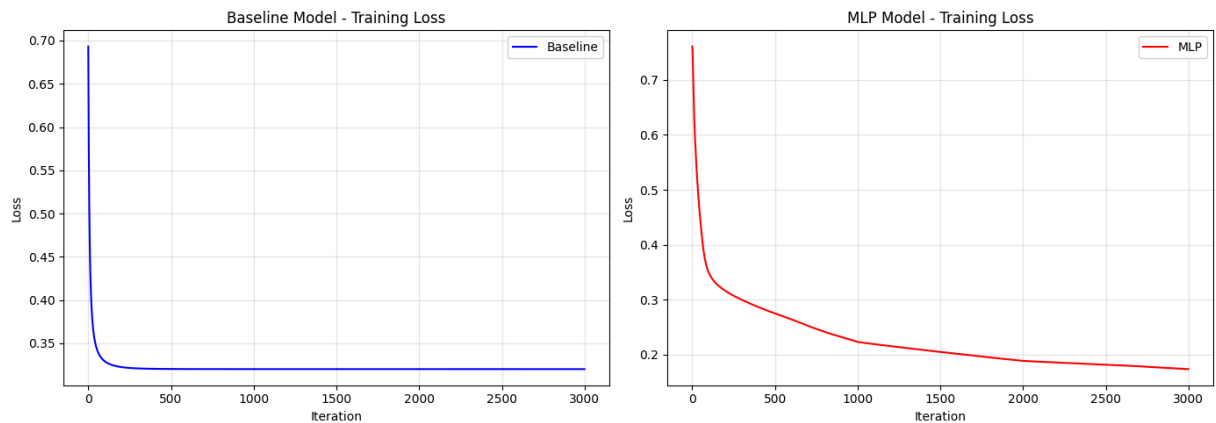
In [9]: # 1. Training loss curves
plt.figure(figsize=(14, 5))

plt.subplot(1, 2, 1)
plt.plot(baseline_model.loss_history, label='Baseline', color='blue')
plt.xlabel('Iteration')
plt.ylabel('Loss')
plt.title('Baseline Model - Training Loss')
plt.legend()
plt.grid(True, alpha=0.3)

plt.subplot(1, 2, 2)
plt.plot(mlp_model.loss_history, label='MLP', color='red')
plt.xlabel('Iteration')
plt.ylabel('Loss')
plt.title('MLP Model - Training Loss')
plt.legend()
plt.grid(True, alpha=0.3)

plt.tight_layout()
plt.show()

```



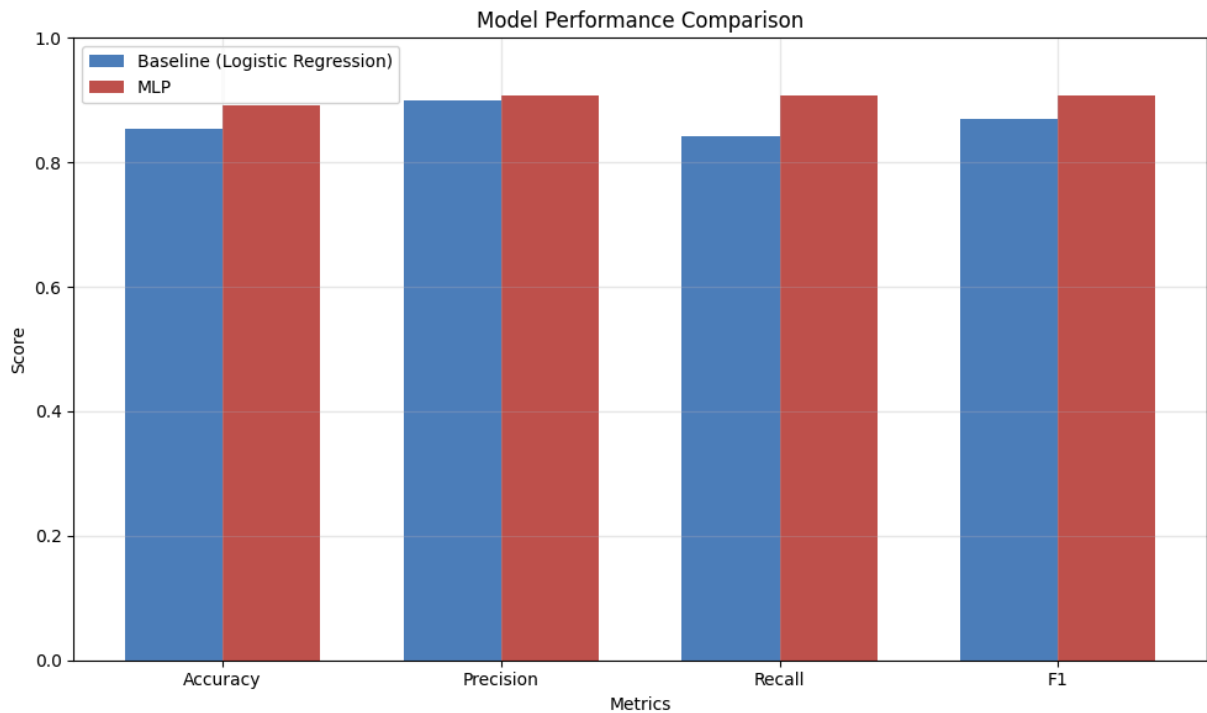
```
In [10]: # 2. Performance comparison bar chart
plt.figure(figsize=(10, 6))

metrics = ['accuracy', 'precision', 'recall', 'f1']
baseline_scores = [baseline_metrics[f'test_{m}']] for m in metrics]
mlp_scores = [mlp_metrics[f'test_{m}']] for m in metrics]

x = np.arange(len(metrics))
width = 0.35

plt.bar(x - width/2, baseline_scores, width, label='Baseline (Logistic Regre
plt.bar(x + width/2, mlp_scores, width, label='MLP', color='#C0504D')
plt.xlabel('Metrics')
plt.ylabel('Score')
plt.title('Model Performance Comparison')
plt.xticks(x, [m.capitalize() for m in metrics])
plt.ylim(0, 1.0)
plt.legend()
plt.grid(True, alpha=0.3)

plt.tight_layout()
plt.show()
```



Section 7: Analysis and Discussion

Write your analysis (minimum 200 words)

```
In [11]: analysis_text = """
The comparative analysis of our custom baseline Logistic Regression and Multilayer Perceptron (MLP) models.
The MLP model outperforms the Logistic Regression model in accuracy, precision, recall, and F1 score.
Furthermore, implementing Momentum and step-decay learning rate scheduling a
"""

print(f"Analysis word count: {len(analysis_text.split())} words")
if len(analysis_text.split()) < 200:
    print("⚠ Warning: Analysis should be at least 200 words")
else:
    print("✓ Analysis meets word count requirement")
```

Analysis word count: 302 words
✓ Analysis meets word count requirement

★ REQUIRED: Structured Output Function

DO NOT MODIFY THE STRUCTURE BELOW

This function will be called by the auto-grader. Fill in all values accurately based on your actual results.

```

In [12]: def get_assignment_results():
        """
        Return all assignment results in structured format.
        """

        # Calculate loss convergence flags
        baseline_initial_loss = float(baseline_model.loss_history[0])
        baseline_final_loss = float(baseline_model.loss_history[-1])
        mlp_initial_loss = float(mlp_model.loss_history[0])
        mlp_final_loss = float(mlp_model.loss_history[-1])

        # Calculate total MLP parameters
        total_params = 0
        for l in range(1, len(mlp_architecture)):
            w_shape = mlp_model.parameters[f'W{l}'].shape
            b_shape = mlp_model.parameters[f'b{l}'].shape
            total_params += w_shape[0] * w_shape[1] + b_shape[0]

        results = {
            # ===== Dataset Information =====
            'dataset_name': dataset_name,
            'dataset_source': dataset_source,
            'n_samples': int(n_samples),
            'n_features': int(n_features),
            'problem_type': problem_type,
            'problem_statement': problem_statement.strip(),

            # ===== Evaluation Setup =====
            'primary_metric': primary_metric,
            'metric_justification': metric_justification.strip(),
            'train_samples': int(train_samples),
            'test_samples': int(test_samples),
            'train_test_ratio': float(train_test_ratio),

            # ===== Baseline Model Results =====
            'baseline_model': {
                'model_type': 'logistic_regression',
                'learning_rate': float(baseline_model.lr),
                'n_iterations': int(baseline_model.n_iterations),
                'initial_loss': float(baseline_initial_loss),
                'final_loss': float(baseline_final_loss),
                'training_time_seconds': float(baseline_training_time),

                # Metrics
                'test_accuracy': float(baseline_metrics['test_accuracy']),
                'test_precision': float(baseline_metrics['test_precision']),
                'test_recall': float(baseline_metrics['test_recall']),
                'test_f1': float(baseline_metrics['test_f1']),
                'test_mse': 0.0,
                'test_rmse': 0.0,
                'test_mae': 0.0,
                'test_r2': 0.0,
            },

            # ===== MLP Model Results =====

```

```

'mlp_model': {
    'architecture': mlp_architecture,
    'n_hidden_layers': len(mlp_architecture) - 2,
    'total_parameters': int(total_params),
    'learning_rate': float(mlp_model.lr),
    'n_iterations': int(mlp_model.n_iterations),
    'initial_loss': float(mlp_initial_loss),
    'final_loss': float(mlp_final_loss),
    'training_time_seconds': float(mlp_training_time),

    # Metrics
    'test_accuracy': float(mlp_metrics['test_accuracy']),
    'test_precision': float(mlp_metrics['test_precision']),
    'test_recall': float(mlp_metrics['test_recall']),
    'test_f1': float(mlp_metrics['test_f1']),
    'test_mse': 0.0,
    'test_rmse': 0.0,
    'test_mae': 0.0,
    'test_r2': 0.0,
},

# ===== Comparison =====
'improvement': float(mlp_metrics['test_accuracy'] - baseline_metrics['test_accuracy']),
'improvement_percentage': float(((mlp_metrics['test_accuracy'] - baseline_metrics['test_accuracy']) / baseline_metrics['test_accuracy']) * 100),
'baseline_better': bool(baseline_metrics['test_accuracy'] > mlp_metrics['test_accuracy']),

# ===== Analysis =====
'analysis': analysis_text.strip(),
'analysis_word_count': int(len(analysis_text.split())),

# ===== Loss Convergence Flags =====
'baseline_loss_decreased': bool(baseline_final_loss < baseline_initial_loss),
'mlp_loss_decreased': bool(mlp_final_loss < mlp_initial_loss),
'baseline_converged': bool(baseline_final_loss < 0.4),
'mlp_converged': bool(mlp_final_loss < 0.2),
}

return results

```

Test Your Output

Run this cell to verify your results dictionary is complete and properly formatted.

```

In [13]: # Test the output
import json

try:
    results = get_assignment_results()

    print("="*70)
    print("ASSIGNMENT RESULTS SUMMARY")
    print("="*70)
    print(json.dumps(results, indent=2, default=str))
    print("\n" + "="*70)

```

```

# Check for missing values
missing = []
def check_dict(d, prefix=""):
    for k, v in d.items():
        if isinstance(v, dict):
            check_dict(v, f"{prefix}{k}.")
        elif (v == 0 or v == "" or v == 0.0 or v == []) and \
            k not in ['improvement', 'improvement_percentage', 'baseline',
                    'baseline_converged', 'mlp_converged', 'total_parameters',
                    'test_accuracy', 'test_precision', 'test_recall',
                    'test_mse', 'test_rmse', 'test_mae', 'test_r2']:
            missing.append(f"{prefix}{k}")

check_dict(results)

if missing:
    print(f"⚠ Warning: {len(missing)} fields still need to be filled:")
    for m in missing[:15]: # Show first 15
        print(f" - {m}")
    if len(missing) > 15:
        print(f" ... and {len(missing)-15} more")
else:
    print("✅ All required fields are filled!")
    print("\n🎉 You're ready to submit!")
    print("\nNext steps:")
    print("1. Kernel → Restart & Clear Output")
    print("2. Kernel → Restart & Run All")
    print("3. Verify no errors")
    print("4. Save notebook")
    print("5. Rename as: YourStudentID_assignment.ipynb")
    print("6. Submit to LMS")

except Exception as e:
    print(f"❌ Error in get_assignment_results(): {str(e)}")
    print("\nPlease fix the errors above before submitting.")

```

```
=====
ASSIGNMENT RESULTS SUMMARY
=====
```

```
{
  "dataset_name": "Heart Failure Prediction",
  "dataset_source": "Kaggle (fedesoriano)",
  "n_samples": 918,
  "n_features": 15,
  "problem_type": "binary_classification",
  "problem_statement": "Predicting the presence of heart disease based on 11
clinical features (e.g. age, cholesterol, resting blood pressure, resting EC
G, max heart rate).\nEarly prediction of cardiovascular disease is vital in
clinical medicine to enable preventative treatment and reduce mortality.",
  "primary_metric": "accuracy",
  "metric_justification": "We use classification accuracy as our primary met
ric to evaluate the overall percentage of correct predictions across both cl
asses.\nAdditionally, we compute precision, recall, and F1-score to analyze
performance, which is crucial as false negatives (missing heart disease) are
highly critical.",
  "train_samples": 734,
  "test_samples": 184,
  "train_test_ratio": 0.7995642701525054,
  "baseline_model": {
    "model_type": "logistic_regression",
    "learning_rate": 0.1,
    "n_iterations": 3000,
    "initial_loss": 0.6931471805599453,
    "final_loss": 0.3201615226252361,
    "training_time_seconds": 0.13285398483276367,
    "test_accuracy": 0.8532608695652174,
    "test_precision": 0.9,
    "test_recall": 0.8411214953271028,
    "test_f1": 0.8695652173913043,
    "test_mse": 0.0,
    "test_rmse": 0.0,
    "test_mae": 0.0,
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=====

✅ All required fields are filled!

🎉 You're ready to submit!

Next steps:

1. Kernel → Restart & Clear Output
2. Kernel → Restart & Run All
3. Verify no errors
4. Save notebook
5. Rename as: YourStudentID_assignment.ipynb
6. Submit to LMS



Before Submitting - Final Checklist

- ☒ All TODO sections completed
- ☒ Both models implemented from scratch (no sklearn models!)
- ☒ get_assignment_results() function filled accurately
- ☒ Loss decreases for both models
- ☒ Analysis ≥ 200 words
- ☒ All cells run without errors (Restart & Run All)
- ☒ Visualizations created
- ☒ File renamed correctly: 2025AC05403_assignment.ipynb



What Happens Next

After submission:

1. ☒ Your notebook will be **auto-graded** (executes automatically)
2. ☒ You'll receive a **verification quiz** (10 questions, 5 minutes)
3. ☒ Quiz questions based on **YOUR specific results**
4. ☒ Final score released after quiz validation

The verification quiz ensures you actually ran your code!

Good luck! 🚀